RL Decision Making for Autonomous Vehicles

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Problem Statement

Conventional autonomous driving systems struggle to handle diverse scenarios, risking catastrophic forgetting. Our solution: a modular approach isolating specific driving situations for targeted training, mitigating forgetting and optimizing performance.

- Traversing a highway
- Navigating roundabouts
- Merging onto the highway

Background

Pre-requisites:

- Understanding reinforcement algorithms

  Familiarity with algorithms like DQN, A2C, A3C. etc.

- Knowledge of autonomous systems

  Understanding the fundamental principles of autonomous systems, including perception, decision-making, and control.

- Background in Machine Learning

  Knowledge of machine learning algorithms, especially neural networks and deep learning frameworks.

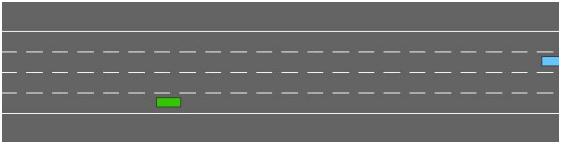
Symbols Used:

- π: Set of policies
- S: State Space
- A: Action space
- s: A state ($s \in S$)
- a: An action ($a \in A$)
- R: Reward function
- r: A reward
- γ: Discount factor for future rewards

Highway-Env Environment

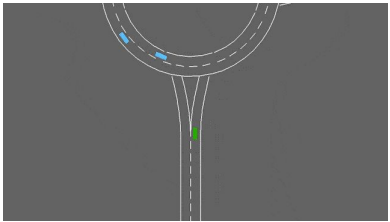
- Collection of environments for decision making in autonomous driving

Highway



Merge

Roundabout



Highway-Env Environment (cont.)

- Observations
  - **Kinematics**, Grayscale Image, Occupancy Grid, Time to Collision
- Rewards

$$R(s,a) = a\frac{v - v_{\min}}{v_{\max} - v_{\min}} - b\,\text{collision}$$

- Actions
  - **Meta Actions**
    - Lane left, lane right, idle, faster, slower
  - **Continuous**
    - low-level controls of the vehicle kinematics namely the throttle (α) and steering angle (δ)

Kinematics Observations

Vehicle	x	y	v_x	v_y
ego-vehicle	5.0	4.0	15.0	0
vehicle 1	-10.0	4.0	12.0	0
vehicle 2	13.0	8.0	13.5	0
…	…	…	…	…
vehicle V	22.2	10.5	18.0	0.5

Chosen Algorithms

Baseline Algorithms:

- Deep Q-Learning (DQN)
- Dueling Deep Q-Learning (Dueling DQN)
- Advantage Actor Critic (A2C)

Improved Algorithms

- Deep Deterministic Policy Gradient (DDPG)
- Proximal Policy Optimization (PPO)

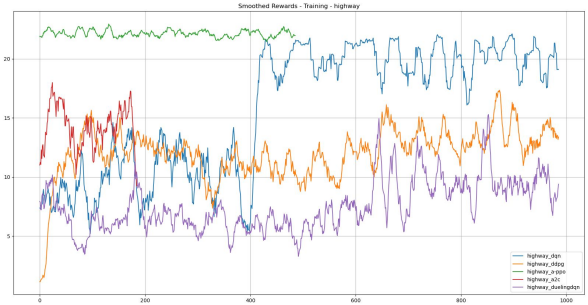
Baseline Algorithms

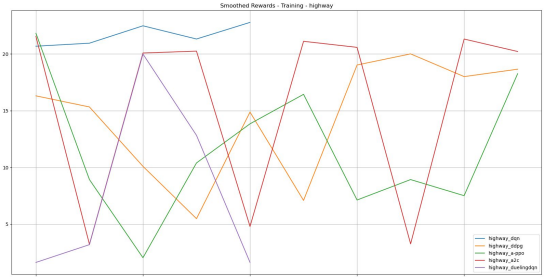
- Deep Q-Learning (DQN)
  - Combines deep learning with Q-learning to learn a policy for making decisions in an environment with discrete actions.
- Dueling Deep Q-Learning (Dueling DQN)
  - An extension of the Deep Q-Network (DQN) algorithm that aims to improve the efficiency and stability of learning by decoupling the estimation of the state value function and the action advantages.
- Advantage Actor Critic (A2C)
  - Combines Policy Gradient Method and Value Based Methods to learn a policy for decision making.

Improved Algorithms

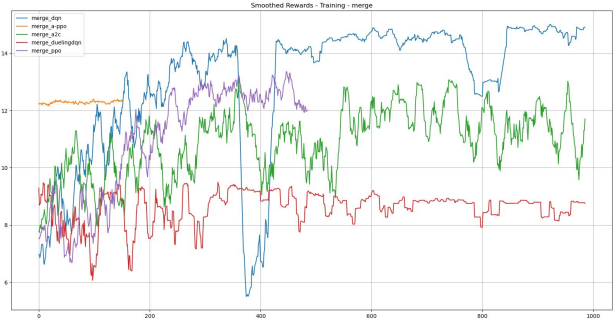
- Deep Deterministic Policy Gradient (DDPG)
  - Uses deep Q-learning and deterministic policy gradients to learn a deterministic policy for continuous action spaces.
- Proximal Policy Optimization (PPO)
  - Fixes some of the limitations of policy gradient methods (Unstability, slow convergence). It does this by adding a surrogate objective function and clipped policy update mechanism.

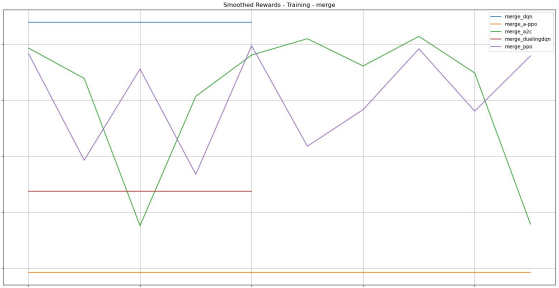
Results- Highway



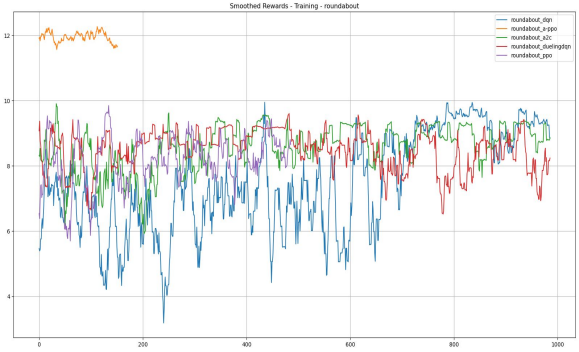


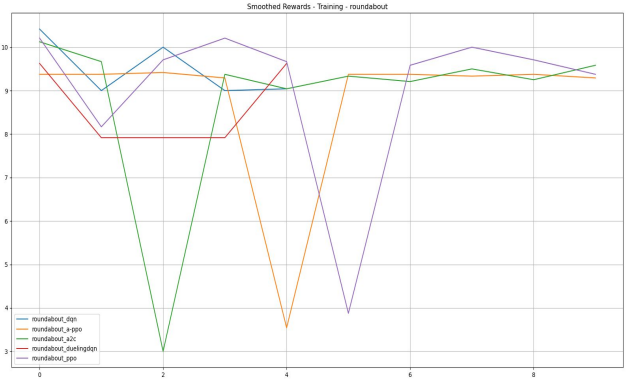
Results - Merge





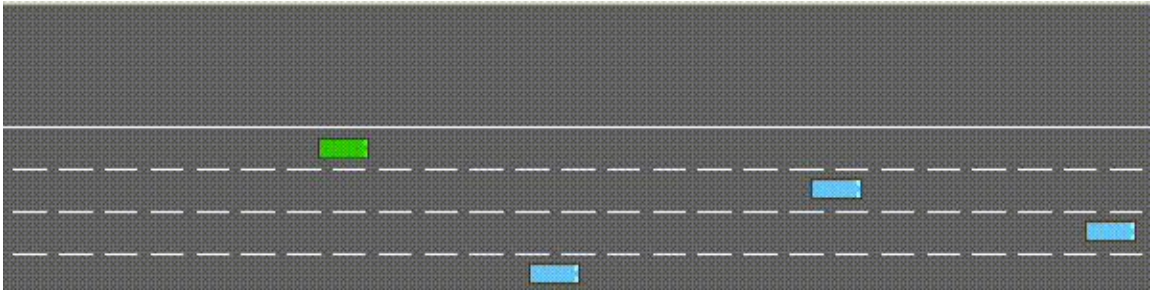
Results - Roundabout





Demo

Preliminary results using DQN, discrete meta actions, and kinematic observations



Key Observations / Summary [1 slide]

After testing out 2 algorithms- DQN and A2C, we realized that A2C performs better to train complex environments such as this.

Future work:

1. Experiment with different algorithms:
    a. Dueling Deep Q-Learning (Dueling DQN)
    b. Deep Deterministic Policy Gradient (DDPG)
    c. Proximal Policy Optimization (PPO)
2. Experiment with different sub-environments:
    a. Lane Changing
    b. Speed Matching
    c. Turning

For Teams: Contribution Summary [1 slide]

Team Member	**Presentation Part**	**Contribution (%)**
Sean Courtney	Problem Statement, Environment, Algorithms	33.33%
Ishaan Malhotra	Background, Key Observations	33.33%
Christian Brazeau	A2C Results, DQN Results	33.33%

Thank you